

- 20** A gene codes for the production of a protein, p53, that binds to damaged DNA during interphase and prevents its replication. A carcinogen in cigarette smoke mutates the gene coding for the p53 protein, preventing production of the protein.

Which statement explains why this mutation may cause cancer?

- A** Cells with no p53 are able to undergo mitosis.
- B** Cells with no p53 replicate their damaged DNA.
- C** The carcinogen in cigarette smoke increases the rate of cell division.
- D** The mutated p53 causes uncontrolled cell division.